

SWITCHING PROJECT CONFIGURATIONS

Wednesday, June 17, 2026 10:26 AM

DC-Switch:-

Hostname Configuration:

-- to configure hostname

```
enable
configure terminal
hostname DC-Switch
```

Trunking Configuration:

-- to configure fa 0/1 -4 as trunk ports

```
interface range fastEthernet 0/1 -4
switchport mode trunk
exit
```

Etherchannel Configuration:

-- to configure etherchannel (LACP)

```
interface range fastEthernet 0/1 -4
channel-group 1 mode passive
exit
```

#####

Core-Switch:-

Hostname Configuration:

-- to configure hostname

```
enable
configure terminal
hostname Core-Switch
```

Trunking Configuration:

-- to configure fa 0/5 -8 as trunk ports for A block

```
interface range fastEthernet 0/5 -8
switchport trunk encapsulation dot1q
switchport mode trunk
exit
```

-- to configure fa 0/9 -12 as trunk ports for B block

```
interface range fastEthernet 0/9 -12
switchport trunk encapsulation dot1q
switchport mode trunk
exit
```

VLAN Trunking Protocol Configuration:

-- to configure VTP Server

```
vtp mode server
vtp version 2
vtp domain ccna.com
vtp password cisco
```

VLAN Configuration:

-- to configure VLANs

```
vlan 10
name HR
exit
```

```
vlan 20
name Sales
exit
```

```
vlan 30
name Marketing
exit
```

```
vlan 40
name IT-Support
exit
```

```
vlan 50
name R&D
exit
```

```
vlan 60
name Operations
exit
```

Etherchannel Configuration:

-- to configure etherchannel (LACP) for Data Centre

```
interface range fastEthernet 0/1 -4
channel-group 1 mode active
exit
```

-- to configure etherchannel (PAGP) for A-Block Switch

```
interface range fastEthernet 0/5 -8
channel-group 2 mode desirable
exit
```

-- to configure etherchannel (PAGP) for B-Block Switch

```
interface range fastEthernet 0/9 -12
channel-group 3 mode desirable
exit
```

Gateway Configuration:

-- to assign IP address to the Port Channel or the Gateway for Data Centre

```
interface port-channel 1
no switchport
ip address 192.168.1.1 255.255.255.0
exit
```

-- to assign IP address to the Gateway for C-Block Router

```
interface fastEthernet 0/13
no switchport
ip address 192.168.2.1 255.255.255.0
Exit
```

Switch Virtual Interface Configuration:

-- to configure SVI for Inter VLAN routing and DORA process

```
interface vlan 10
ip address 10.0.0.1 255.0.0.0
ip helper-address 192.168.1.20
no shutdown
exit
```

```
interface vlan 20
ip address 20.0.0.1 255.0.0.0
ip helper-address 192.168.1.20
no shutdown
exit
```

```
interface vlan 30
ip address 30.0.0.1 255.0.0.0
ip helper-address 192.168.1.20
no shutdown
exit
```

```
interface vlan 40
ip address 40.0.0.1 255.0.0.0
ip helper-address 192.168.1.20
no shutdown
exit
```

```
interface vlan 50
ip address 50.0.0.1 255.0.0.0
ip helper-address 192.168.1.20
no shutdown
exit
```

```
interface vlan 60
ip address 60.0.0.1 255.0.0.0
ip helper-address 192.168.1.20
no shutdown
exit
```

Inter VLAN Routing Configuration:

-- to perform Inter VLAN Routing

```
ip routing
```

Default Route Configuration:

-- to add a default route for DHCP

```
ip route 0.0.0.0 0.0.0.0 192.168.2.2
```

#####

A-Block-Switch:-

Hostname Configuration:

-- to configure hostname

```
enable
configure terminal
hostname A-Block-Switch
```

Trunking Configuration:

-- to configure fa 0/1 -4 as trunk ports

```
interface range fastEthernet 0/1 -4
switchport mode trunk
exit
```

VTP Configuration:

-- to configure VTP Client

```
vtp mode client
vtp version 2
vtp domain ccna.com
vtp password cisco
```

VLAN Configuration:

-- to assign the respective ports to the VLANs

```
interface range fastEthernet 0/5 -6
switchport access vlan 10
exit
```

```
interface range fastEthernet 0/7 -8
switchport access vlan 20
exit
```

```
interface range fastEthernet 0/9 -11
switchport access vlan 30
exit
```

Etherchannel Configuration:

-- to configure etherchannel (PAGP)

```
interface range fastEthernet 0/1 -4
channel-group 2 mode auto
exit
```

#####

B-Block-Switch:-

Hostname Configuration:

-- to configure hostname

```
enable
```

```
configure terminal
hostname B-Block-Switch
```

Trunking Configuration:

-- to configure fa 0/1 -4 as trunk ports

```
interface range fastEthernet 0/1 -4
switchport mode trunk
Exit
```

VTP Configuration:

-- to configure VTP Client

```
vtp mode client
vtp version 2
vtp domain ccna.com
vtp password cisco
```

VLAN Configuration:

-- to assign the respective ports to the VLANs

```
interface range fastEthernet 0/5 -6
switchport access vlan 40
exit
```

```
interface range fastEthernet 0/7 -8
switchport access vlan 50
exit
```

```
interface range fastEthernet 0/9 -10
switchport access vlan 60
exit
```

Etherchannel Configuration:

-- to configure etherchannel (PAGP)

```
interface range fastEthernet 0/1 -4
channel-group 3 mode auto
exit
```

#####

C-Block-Switch:-

Hostname Configuration:

-- to configure hostname

```
enable
configure terminal
hostname C-Block-Switch
```

Trunking Configuration:

-- to configure fa/1 as trunk port

```
interface fastEthernet 0/1
switchport mode trunk
exit
```

VLAN Configuration:

-- to configure VLANs

```
vlan 70
name Finance
exit
```

```
vlan 80
name Legal
exit
```

```
vlan 90
name Management
exit
```

-- to assign respective ports to the VLANs

```
interface range fastEthernet 0/2 -3
switchport access vlan 70
exit
```

```
interface fastEthernet 0/4
switchport access vlan 80
exit
```

```
interface range fastEthernet 0/5 -6
switchport access vlan 90
exit
```

#####

C-Block-Router:-

Hostname Configuration:

-- to configure hostname

```
enable
configure terminal
hostname C-Block-Router
```

Gateway Configuration:

-- to configure fa 0/0 port

```
interface fastEthernet 0/0
ip address 192.168.2.2 255.255.255.0
no shutdown
exit
```

Router On A Stick Configuration:

-- to configure sub interfaces for the VLANs

```
interface fastEthernet 0/1.1
encapsulation dot1q 70
ip address 70.0.0.1 255.0.0.0
ip helper-address 192.168.1.20
no shutdown
exit
```

```
interface fastEthernet 0/1.2
encapsulation dot1q 80
ip address 80.0.0.1 255.0.0.0
ip helper-address 192.168.1.20
no shutdown
exit
```

```
interface fastEthernet 0/1.3
encapsulation dot1q 90
ip address 90.0.0.1 255.0.0.0
ip helper-address 192.168.1.20
no shutdown
exit
```

Inter VLAN Routing Configuration:

-- to up the sub-interfaces

```
interface fastEthernet 0/1
no shutdown
```

Default Route Configuration:

-- to add a default route for DHCP

```
ip route 0.0.0.0 0.0.0.0 192.168.2.1
```